

Generative AI Uses and Challenges In Kashmir

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A Big Digital Shift - Introduction

For many decades, Jammu and Kashmir faced numerous challenges due to its difficult mountainous terrain, harsh weather conditions, and complex political situation, which often made governance and public service delivery slow and inefficient. One of the most notable examples of this was the historic Darbar Move, a 150-year-old administrative practice in which the government shifted its offices between Srinagar and Jammu every six months. During this process, thousands of official documents and records were physically transported in trucks across the region, requiring significant time, manpower, and financial resources. However, with the rapid advancement and development of technology and the government's commitment to modernization, Jammu and Kashmir has entered a new era of digital governance which is very evident and very convenient for the people. There was also a case where someone had Bad road condition in their colony and they registered a grievance about it their issue was solved within some days.

The traditional Darbar Move has been discontinued, and its place has been taken by a modern digital system where official records are stored electronically and can be accessed instantly from any location. This transformation has greatly improved administrative efficiency, reduced costs, and minimized delays in decision-making. Today, citizens can access more than 1,100 government services online through digital platforms, allowing them to apply for certificates, licenses, permits, welfare schemes, and various public services without the need to visit government offices or wait in long queues, if someone has any complaints or suggestions they can use the E-Samarth Portal provided by the Govt. of Jammu and Kashmir , where they just have to provide the proof and every problem they are facing and concerned authorities will fix their issues in a specified time .

accessible even to people living in remote and rural areas. This remarkable shift from paper-based administration to digital governance

Saving the Kashmiri Language

For many years, the Kashmiri language faced a unique challenge in the digital age. And in Kashmir many families prefer that their children speak in languages like English and Urdu , as many people consider kashmiri to be very outdates , while some people continue to speak and write Kashmiri in their daily lives, computers and modern software often struggled to understand the language correctly.

One major reason was the absence of diacritical marks—special symbols that indicate the correct pronunciation and meaning of words. Without these marks, digital text could become confusing, making it difficult for translation systems, speech-recognition software, and language-processing tools to interpret Kashmiri accurately. As a result, the language remained underrepresented in many modern technologies, limiting its presence on digital platforms and reducing opportunities for its growth in the online world.

To address this challenge, a team of dedicated local engineers and language experts developed an innovative artificial intelligence tool known as the "Koshur Diacritizer." This advanced system automatically analyzes Kashmiri text and inserts the missing pronunciation marks with a high level of accuracy. What once required significant manual effort can now be completed within seconds using AI. The development of this tool represents an important step toward preserving and modernizing the Kashmiri language. By enabling computers to better understand written Kashmiri, the Koshur Diacritizer opens the door for more accurate translation applications, smarter voice assistants, improved text-to-speech systems, and educational technologies designed specifically for Kashmiri speakers.

Beyond its technical achievements, the project carries deep cultural significance. Language is an essential part of a community's identity, history, and heritage, and ensuring that Kashmiri can thrive in the digital era helps protect that heritage for future generations. Young people will be able to interact with their native language through modern devices and online

platforms, while researchers, educators, and developers can create new digital resources more easily. **The Koshur Diacritizer (developed by Haq Nawaz Malik, Nahfid Nissar, Faizan Iqbal)** demonstrates how technology can be used not only to solve practical problems but also to preserve cultural traditions and strengthen linguistic diversity. By bringing Kashmiri into the world of artificial intelligence and digital communication, this innovation ensures that the language remains vibrant, accessible, and relevant in an increasingly technology-driven future.

As language is the most important part which keeps a society together such interventions and inventions will Kashmiri language to sustain in a longer term and also teach the younger generation about Kashmiri language There have also been many people who have made tools which can save handicraft or give the artisans more leverage.

To keep it all together we can say that generative AI has a very great role in overall technology development as well as the development of infrastructure in Kashmir. With regards to language as well as other things which we'll discuss.

Smarter Farming and Crafts

Agriculture and its related fields have always been/acted as the backbone of Jammu and Kashmir's economy, providing livelihoods to thousands of families across the region. From the vast apple orchards of Kashmir Valley to the world-famous saffron fields of Pampore, farming plays a vital role in both local culture and economic development. However, farmers have traditionally faced many challenges, including unpredictable weather, plant diseases, pest attacks, and difficulties in maintaining product quality. Talking of today, as there are many advancements in Artificial Intelligence (AI) and digital technology are helping farmers overcome these obstacles and making agriculture smarter, more efficient, and more profitable. In order to make it very efficient and very helpful there are a lot of applications of Generative AI for us. Agricultural and its related fields have always been a main thing in terms of providing employment to the people of Jammu and Kashmir so with the help of Generative AI it will improve the situation of people.

One of the most significant changes can be seen in Kashmir's apple industry, which is among the largest horticultural sectors in India. AI-powered cameras and computer vision systems are now being used in orchards to monitor the health of apple trees. These intelligent systems can detect early signs of diseases, nutrient deficiencies, and pest infestations that may not be easily visible to the human eye. By identifying problems at an early stage, farmers can take timely action and prevent large-scale crop losses. AI is also being used in apple grading and sorting facilities, where advanced cameras examine each fruit for size, color, shape, and quality. Apples can be sorted automatically without causing damage, ensuring that only the best-quality produce reaches markets while reducing waste and increasing profits for growers.

Technology is also transforming the cultivation and protection of saffron, often called the world's most expensive spice. Saffron from Kashmir is renowned for its quality and aroma, but the market has long struggled with the problem of counterfeit products. Many consumers find it difficult to

distinguish genuine saffron from cheaper imitations. To solve this issue, scientists and researchers have developed a simple AI-powered smartphone application that can quickly analyze saffron samples and determine their authenticity. Farmers, traders, and buyers can use this technology to verify products with greater confidence, helping protect the reputation of Kashmiri saffron in national and international markets. This innovation not only safeguards consumers but also ensures that genuine saffron growers receive fair value for their hard work.

Beyond agriculture and other things , Artificial Intelligence is helping preserve and strengthen Kashmir's rich artistic heritage. The region is famous worldwide for its handwoven carpets, which are known for their intricate patterns, exceptional craftsmanship, and cultural significance. For centuries, carpet weaving has relied on a traditional coding system called Talim, a unique language of symbols and instructions that guides weavers in creating complex designs. Learning, reading, and preparing Talim codes has historically been a time-consuming process requiring specialized expertise. In some cases, preparing the codes for a single carpet design could take several months.

Today, AI-based software is revolutionizing this process by reading, interpreting, and digitizing Talim codes much more quickly and accurately. What once required months of painstaking work can now be completed in a matter of weeks. Digital preservation of Talim ensures that this centuries-old knowledge is not lost as older generations retire and fewer people learn the traditional methods. At the same time, it enables weavers to access designs more easily, improve productivity, and create high-quality carpets at a faster pace. By combining ancient craftsmanship with modern technology, Kashmir is finding innovative ways to preserve its cultural heritage while creating new economic opportunities for artisans.

Better Healthcare and Tourism

Healthcare and tourism are two sectors that have a direct impact on the lives of people in Jammu and Kashmir. While the region is known around the world for its breathtaking landscapes, snow-capped mountains, beautiful lakes, and rich cultural heritage, it has also faced challenges in providing quality healthcare to people living in remote and mountainous areas. Today, Artificial Intelligence (AI) and digital technologies are beginning to transform both sectors, creating new opportunities to improve people's lives while supporting economic growth and development.

In the field of healthcare, access to medical services has long been a challenge for many residents, particularly those living in isolated villages surrounded by mountains and difficult terrain. During harsh winters especially in Kashmir , heavy snowfall can block roads and make travel to hospitals extremely difficult. In some cases, patients must travel several hours simply to consult a specialist or receive a diagnosis. Recognizing these challenges, healthcare institutions and technology innovators are increasingly exploring the use of Artificial Intelligence to bridge the gap between doctors and patients.

Major healthcare institutions, including AIIMS Jammu, are planning to integrate AI-powered tools that can assist doctors in diagnosing illnesses more quickly and accurately. These systems are designed to analyze medical data, identify patterns that may be difficult for humans to detect, and provide valuable support during diagnosis. For example, AI can help examine medical images such as X-rays, CT scans, and MRIs, enabling healthcare professionals to detect diseases at earlier stages. Early diagnosis is often critical because it allows treatment to begin sooner, increasing the chances of recovery and reducing healthcare costs for patients and families. Rather than replacing doctors, these technologies act as intelligent assistants that help medical professionals make better-informed decisions and deliver higher-quality care.

At the same time, local students, researchers, and young entrepreneurs are contributing innovative solutions to healthcare challenges through the development of AI-powered telehealth applications. These mobile platforms are designed specifically to help people living in distant mountain villages where access to hospitals and specialists may be limited. Through a smartphone, residents can connect with healthcare professionals, describe their symptoms, receive preliminary medical guidance, and determine whether they need immediate medical attention. Such systems can be particularly valuable for elderly patients, individuals with mobility difficulties, and families living far from urban centers. By reducing the need for long and expensive journeys, telehealth services can make healthcare more accessible, affordable, and convenient for thousands of people.

Tourism, another vital pillar of Jammu and Kashmir's economy, is also being transformed by modern technology. Every year, millions of visitors travel to the region to experience destinations such as Srinagar, Gulmarg, Pahalgam, Sonamarg, and many other scenic locations. While tourism generates employment and supports local businesses, it also places significant pressure on transportation systems, public services, and infrastructure, particularly during peak travel seasons. Traffic congestion in popular tourist areas can lead to delays, pollution, and frustration for both residents and visitors.

To address these challenges, authorities are increasingly using AI-powered traffic management systems and smart cameras to monitor vehicle movement and improve traffic flow. These intelligent systems can analyze road conditions in real time, identify congestion points, and help officials make faster decisions to reduce traffic jams. By optimizing traffic signals and providing accurate information to authorities, AI can contribute to smoother transportation networks and a more pleasant experience for tourists. Visitors can spend less time stuck in traffic and more time enjoying the natural beauty and cultural attractions of the region.

However, despite the many advantages of Artificial Intelligence, local leaders and policymakers remain mindful of its social impact. Tourism is one of the largest sources of employment in Jammu and Kashmir, supporting

thousands of hotel workers, tour guides, taxi drivers, restaurant employees, artisans, and small business owners. For many families, tourism-related jobs are a primary source of income. As a result, there is a strong emphasis on ensuring that technology complements human workers rather than replacing them. While AI can help automate routine tasks and improve efficiency, the warmth, hospitality, cultural knowledge, and personal interactions provided by local people remain irreplaceable aspects of the visitor experience.

The goal is therefore to create a balanced approach in which technology enhances services while preserving employment opportunities. A tourist may benefit from smarter transportation systems and digital assistance, but they will still rely on local guides to share stories, hotel staff to provide hospitality, and artisans to showcase traditional crafts. In this way, technology becomes a tool that supports people rather than a substitute for them.

The Dark Side: Fake News and the Digital Divide

While Artificial Intelligence is bringing some opportunities to Jammu and Kashmir, it is also very important to recognize that technological progress also comes with serious challenges. Like any powerful tool, AI can be used for both positive and negative purposes. As the region becomes more connected and digitally advanced, new risks are emerging that require careful attention from governments, technology companies, educators, and ordinary citizens. Among the most pressing concerns are the spread of misinformation through AI-generated content and the growing digital divide between urban and rural communities.

One of the most alarming challenges is the rise of **deepfakes**, which are highly realistic but completely fake videos, audio recordings, and images created using Artificial Intelligence. These technologies have become so advanced that it is often difficult for ordinary people to distinguish between what is real and what has been digitally manipulated. A deepfake can make it appear as though a political leader, government official, military officer, or public figure said or did something that never actually happened.

In a region as sensitive as Jammu and Kashmir, where information can have a significant impact on public opinion and social stability, the misuse of deepfake technology poses a serious threat. False videos or fabricated audio clips can spread rapidly across social media platforms, messaging applications, and online communities within a matter of minutes. Before authorities have an opportunity to verify the information, thousands of people may have already viewed, shared, and believed it. Such misinformation can create confusion, fear, panic, and mistrust among communities.

Imagine a situation where a fake video appears to show an army officer making a controversial statement or a political leader announcing false information during a tense situation. Even if the content is eventually proven false, the damage may already have been done. Rumors can spread quickly, public confidence can be shaken, and unnecessary anxiety can affect entire

communities. This is why cybersecurity experts, government agencies, and media organizations are increasingly focusing on methods to detect AI-generated misinformation and educate citizens about verifying information before sharing it online.

The challenge is not only technical but also social. People naturally tend to trust what they see and hear. For generations, photographs, videos, and audio recordings were considered strong evidence of reality. AI has changed that assumption. In today's digital world, seeing is no longer always believing. As a result, digital literacy—the ability to critically evaluate information online—has become just as important as access to technology itself. Citizens need to learn how to identify trustworthy sources, verify news reports, and approach sensational content with caution. Building awareness is one of the strongest defenses against misinformation.

However, even as some people worry about the dangers of advanced technology, many others face a completely different challenge: they are still struggling to access that technology at all. This issue is commonly known as the **digital divide**, and it remains one of the biggest obstacles to inclusive development in Jammu and Kashmir.

In major urban centers such as Srinagar and Jammu, many residents have access to smartphones, broadband internet, digital services, and online learning platforms. Students can attend virtual classes, businesses can operate online, and citizens can access government services with just a few clicks. For these communities, digital technology is becoming an increasingly important part of everyday life.

The situation is often very different in many remote villages scattered across the mountains and valleys of the region. In these areas, residents may experience frequent electricity outages, weak internet connectivity, limited access to digital devices, and a lack of technical training. For some families, purchasing computers or smartphones remains financially challenging. Others may have access to devices but lack the digital skills needed to use them effectively.

This gap creates inequality in access to opportunities. A student in a city with reliable internet can benefit from online educational resources, digital libraries, and AI-powered learning tools, while a student in a remote village may struggle to participate in online classes due to poor connectivity. Similarly, farmers who could benefit from AI-driven agricultural advice may be unable to access these services if they lack internet access or digital literacy. Healthcare applications, government portals, and online business opportunities are only useful when people can reliably connect to them.

The digital divide is not merely a technological problem—it is a social and economic issue that affects education, healthcare, employment, and overall quality of life. If left unaddressed, it risks creating a situation where the benefits of Artificial Intelligence and digital transformation are concentrated among a limited portion of the population while others are left behind. For technological progress to be truly meaningful, it must reach every community, regardless of geography or economic status.

Addressing this challenge requires investment in infrastructure, including reliable electricity, stronger internet networks, and affordable digital devices. It also requires training programs that help people develop the skills needed to use technology confidently and safely. Schools, community centers, and local organizations can play an important role in teaching digital literacy and ensuring that citizens understand both the opportunities and risks associated with online technologies.

Ultimately, the future of Artificial Intelligence in Jammu and Kashmir will depend not only on technological innovation but also on inclusiveness and responsibility. AI has the potential to improve healthcare, strengthen agriculture, preserve cultural heritage, enhance governance, and boost economic growth. However, these benefits can only be fully realized if society is prepared to address the challenges that accompany them. Combating misinformation, protecting people from digital manipulation, expanding internet access, and improving digital literacy are all essential steps toward building a safer and more equitable digital future.

Conclusion

The rapid growth of Artificial Intelligence and digital technologies is transforming Jammu and Kashmir in ways that were unimaginable just a few decades ago. The successful digitization of government services has made public administration more transparent, accessible, and responsive, while AI-powered innovations are helping solve challenges that are unique to the geographical and socio-economic conditions of Jammu and Kashmir.

One of the most significant achievements is the use of Artificial Intelligence to preserve and promote the Kashmiri language. Projects such as the Koshur Diacritizer demonstrate how modern technology can help protect cultural identity and ensure that local languages remain relevant in the digital age. Similarly, AI applications in apple orchards, saffron cultivation, and traditional handicrafts show that technological progress does not necessarily replace tradition; instead, it can strengthen and modernize it while creating new economic opportunities.

The healthcare and tourism sectors are also benefiting from AI-driven solutions. Improved diagnostics, telemedicine services, smart traffic management systems, and better tourism planning have the potential to enhance the quality of life for residents and improve the overall experience for visitors.

At the same time, the digital divide between urban and rural communities threatens to exclude many people from the benefits of technological advancement. Issues such as limited internet access, inadequate digital literacy, and unequal access to digital resources for many people.

At this point Kashmir only needs more students who are into things like AI computer science so that we have a good team of people who will develop more solutions with great security and ethical use that can benefit the people of Kashmir as well as it can benefit the students who are the upcoming generation of the Valley.

REFERENCES

Government of Jammu and Kashmir. (2024). Digital J&K: E-Governance and Digital Transformation Initiatives. Srinagar/Jammu: Government of Jammu and Kashmir.

Government of Jammu and Kashmir. (2024). Artificial Intelligence Policy Framework for Jammu and Kashmir. Department of Information Technology, Government of Jammu and Kashmir.

National Informatics Centre (NIC). (2024). E-UNNAT Portal and Digital Service Delivery in Jammu & Kashmir. New Delhi: Ministry of Electronics and Information Technology.

Sher-e-Kashmir University of Agricultural Sciences and Technology of Kashmir (SKUAST-K). (2024). Applications of Artificial Intelligence in Apple Orchard Management and Precision Agriculture. Srinagar: SKUAST-K.

Sher-e-Kashmir University of Agricultural Sciences and Technology of Kashmir (SKUAST-K). (2024). Research Initiatives in Saffron Authentication and Yield Optimization Using Artificial Intelligence. Srinagar: SKUAST-K.

Centre for Development of Advanced Computing (C-DAC). (2024). AI-Based Fruit Grading and Quality Assessment Technologies. Kolkata: Ministry of Electronics and Information Technology.

Malik, H. N., Nissar, N., & Iqbal, F. (2024). Koshur Diacritizer: Automatic Diacritic Restoration for Kashmiri Language Processing. Kashmir AI Research Publications.

Kashmir AI Research. (2024). Low-Resource Natural Language Processing for Kashmiri: Translation, Classification and Language Preservation. Srinagar: Kashmir AI Research.

Google Research. (2022). ByT5: Towards a Token-Free Future with Pre-trained Byte-to-Byte Models. Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP).

All India Institute of Medical Sciences (AIIMS), Jammu. (2024). Proposal for the Global Centre of Excellence for Artificial Intelligence in Healthcare. Jammu: AIIMS Jammu.

World Health Organization. (2023). Ethics and Governance of Artificial Intelligence for Health. Geneva: WHO.

International Telecommunication Union. (2023). Measuring Digital Development: Facts and Figures 2023. Geneva: ITU.

Press Information Bureau (PIB). (2024). Fact Check Reports on AI-Generated Misinformation and Deepfakes in India. New Delhi: Government of India.